SELF-STUDY TRACK — FULL TIME

AI-native engineer roadmap

A 16-week plan for transitioning from Java / Spring Boot / Angular full-stack development into AI-native engineering — LLM systems, RAG, agents, and evaluation — with a portfolio and freelance track built in. Compressed from an 18-week source curriculum, reordered to weight time toward the highest-leverage material for shipping products rather than research.

16	16	16	16
WEEKS	PROJECTS	PAPERS	BUILD LOGS

HOW THIS IS SEQUENCED

- **Weeks 1–3, foundations** — math and classical ML at fluency level, not textbook depth. You already have engineering intuition; this is about recognizing the vocabulary.
- **Weeks 4–6, transformers** — the one place worth not rushing. Attention, KV cache, RoPE, quantization, LoRA — build GPT from scratch here.
- **Week 7, systems** — fast pass, since backend engineering (FastAPI, Postgres, Redis) largely maps onto your existing Spring Boot experience.
- **Weeks 8–12, the core** — LLM engineering, RAG, agents, evaluation, and security. This is where freelance-billable skill actually lives. Go slow.
- **Weeks 13–14, post-training** — conceptual pass on RLHF/DPO/GRPO. Deprioritize deep implementation unless targeting a research role.
- **Weeks 15–16, capstone** — one full deployed system with an eval harness, then repeat the pattern on 2–3 smaller projects to build the freelance portfolio.

> **Weekly rhythm, every week without exception:** study the topic → build the project → read + critique the paper → write one LinkedIn/X post about what you built or learned. The post is not optional — it's what turns 16 weeks of study into a visible portfolio and inbound freelance interest.

BLOCK A — FOUNDATIONS (COMPRESSED)

W01

Probability, statistics & information theory

STUDY Bayes' rule, MLE, entropy, cross-entropy, KL divergence — fluency, not textbook mastery.

PROJECT Bayesian spam classifier from scratch in NumPy.

PAPER A Few Useful Things to Know About Machine Learning — Pedro Domingos

☐ STUDY ☐ PROJECT ☐ PAPER ☐ LINKEDIN/X POST

W02

Linear algebra, calculus & optimization

STUDY Vectors, SVD, chain rule, gradient descent, Adam — the math behind backprop.

PROJECT Build a scalar autograd engine (micrograd-style) + visualize SGD vs Adam on a loss surface.

PAPER micrograd walkthrough — Andrej Karpathy

☐ STUDY ☐ PROJECT ☐ PAPER ☐ LINKEDIN/X POST

W03

Classical ML + deep learning foundations

STUDY Logistic regression, regularization, MLPs, backprop, BatchNorm vs LayerNorm.

PROJECT Logistic regression from scratch, then a full MLP trained on MNIST (target >98%).

PAPER Gradient-Based Learning Applied to Document Recognition — LeCun et al. (1998)

☐ STUDY ☐ PROJECT ☐ PAPER ☐ LINKEDIN/X POST

BLOCK B — TRANSFORMERS + LLM INTERNALS

W04

Sequence models, attention & tokenization

STUDY RNNs, LSTMs, the birth of attention, BPE tokenization, embeddings.

PROJECT Char-level LSTM language model + a BPE tokenizer built from scratch.

PAPER Neural Machine Translation by Jointly Learning to Align and Translate — Bahdanau et al.

☐ STUDY ☐ PROJECT ☐ PAPER ☐ LINKEDIN/X POST

W05

Transformer architecture deep dive

STUDY Self-attention, multi-head attention, RoPE, SwiGLU, GQA, KV cache — go slow here.

PROJECT Build GPT from scratch in PyTorch, train on Tiny Shakespeare until output is coherent.

PAPER Attention Is All You Need — Vaswani et al.

☐ STUDY ☐ PROJECT ☐ PAPER ☐ LINKEDIN/X POST

W06

Inference, quantization & fine-tuning mechanics

STUDY Sampling strategies, quantization (INT8/4-bit), LoRA/QLoRA, vLLM basics.

PROJECT Deploy an open-source LLM with vLLM, then LoRA fine-tune it on a custom dataset.

PAPER LoRA: Low-Rank Adaptation of Large Language Models — Hu et al.

☐ STUDY ☐ PROJECT ☐ PAPER ☐ LINKEDIN/X POST

BLOCK C — SYSTEMS (FAST PASS — LEVERAGES YOUR BACKEND BACKGROUND)

W07 Backend for AI products + data engineering essentials

STUDY FastAPI streaming, Postgres/Redis, then vector DB internals (HNSW), chunking, embeddings.

PROJECT FastAPI streaming LLM API with WebSockets + a chunking/embedding pipeline into Pgvector.

PAPER Designing Data-Intensive Applications, Ch.1 — Kleppmann

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

BLOCK D — LLM ENGINEERING, RAG & AGENTS (CORE, GO SLOW)

W08 LLM engineering & context engineering

STUDY Prompting patterns, context window design, structured outputs, memory systems.

PROJECT Perplexity-style clone: streaming answers with citations and source attribution.

PAPER Chain-of-Thought Prompting Elicits Reasoning in Large Language Models — Wei et al.

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

W09 Retrieval science & advanced RAG

STUDY BM25, dense retrieval, hybrid fusion, HyDE, reranking, chunking science.

PROJECT Hybrid BM25 + dense RAG pipeline; benchmark naive vs HyDE vs reranked retrieval.

PAPER HyDE: Precise Zero-Shot Dense Retrieval without Relevance Labels — Gao et al.

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

W10 Agent engineering

STUDY ReAct loop, planner-executor, multi-agent patterns, human-in-the-loop design.

PROJECT Coding agent with tool use (file read/write, bash, search) + human approval on destructive actions.

PAPER ReAct: Synergizing Reasoning and Acting in Language Models — Yao et al.

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

BLOCK E — EVALUATION, RELIABILITY & SECURITY

W11 Evaluation & reliability engineering

STUDY Golden datasets, LLM-as-judge, tracing, cost tracking, canary deployments.

PROJECT Full eval harness for your Week 10 agent: LLM-as-judge, cost dashboard, tracing.

PAPER Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference — Zheng et al.

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

W12 AI security & red teaming

STUDY Prompt injection, jailbreaking, RAG poisoning, permission boundaries, red team method.

PROJECT Red-team your own agent — find and document 5+ attacks, then patch each one.

PAPER Universal and Transferable Adversarial Attacks on Aligned Language Models — Zou et al.

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

BLOCK F — POST-TRAINING (LIGHT PASS — TRIM UNLESS RESEARCH-TRACK)

W13

RLHF, DPO & alignment concepts

STUDY SFT, reward models, RLHF pipeline, the DPO simplification — know it for interviews.

PROJECT DPO fine-tune on a small preference dataset; compare base vs SFT vs DPO output.

PAPER Direct Preference Optimization — Rafailov et al.

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

W14

Reasoning models & frontier research

STUDY Test-time compute, process vs outcome reward models, GRPO — conceptual pass.

PROJECT Light: best-of-N sampling experiment on a small reasoning benchmark. Skip full GRPO implementation unless targeting a research role.

PAPER DeepSeek R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

BLOCK G — CAPSTONE & HIRING SPRINT

W15

Capstone — system design & build

STUDY Pick a track (AI SWE, FDE, Applied AI, or AI Infra). Write the system design doc and ADRs before coding.

PROJECT Start building: full-stack, deployed, not localhost. Frontend + backend + DB + agent.

PAPER LLM-as-a-Judge — Zheng et al. (use for your capstone eval harness)

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

W16

Capstone — ship, evaluate & demo

STUDY Eval harness, monitoring dashboard, postmortem, then resume/portfolio audit and outreach.

PROJECT Deploy, add eval + cost dashboard, write the postmortem, record a 20-minute live demo.

PAPER Re-read one paper most relevant to your capstone domain and cite it in your write-up.

▯ STUDY ▯ PROJECT ▯ PAPER ☐ LINKEDIN/X POST

Freelance & OSS tracks, running throughout: once you have your first deployed project (around week 8), start logging outreach — Upwork, warm network, cold DMs referencing your build-in-public posts. Target one open-source PR per month to LangGraph, vLLM, LiteLLM, OpenHands, LlamaIndex, DSPy, PydanticAI, Haystack, or CrewAI.

Prepared as a companion to the interactive tracker. Timeline assumes full-time commitment; expect drift on the harder architecture and systems weeks, and that's fine — the sequencing matters more than the exact week count.